

UNIT-6: ANALOGIES AMONG MOMENTUM, ENERGY & MASS TRANSPORT

① Molecular Transport of Momentum

Newton's Law of Viscosity

$$\tau_{yx} = -\mu \frac{dv_x}{dy}$$

①

Momentum diffusivity, $\nu = \frac{\mu}{\rho}$

② Molecular Transport of Energy

Fourier's Law of Heat conduction

$$q = -k \frac{dT}{dy}$$

②

Thermal diffusivity, $\alpha = \frac{k}{\rho \hat{C}_p}$

③ Molecular Transport of Mass

Fick's Law of Diffusion

$$J'_A = -D_{AB} \frac{dC_A}{dy}$$

③

D_{AB} = Mass diffusivity

Equations ①, ② & ③ have similar form. Therefore, the molecular transport of momentum, energy & mass is analogous.

Prandtl Number

$$Pr = \frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}} = \frac{\nu}{\alpha} = \frac{\mu \hat{C}_p}{k}$$

Schmidt Number

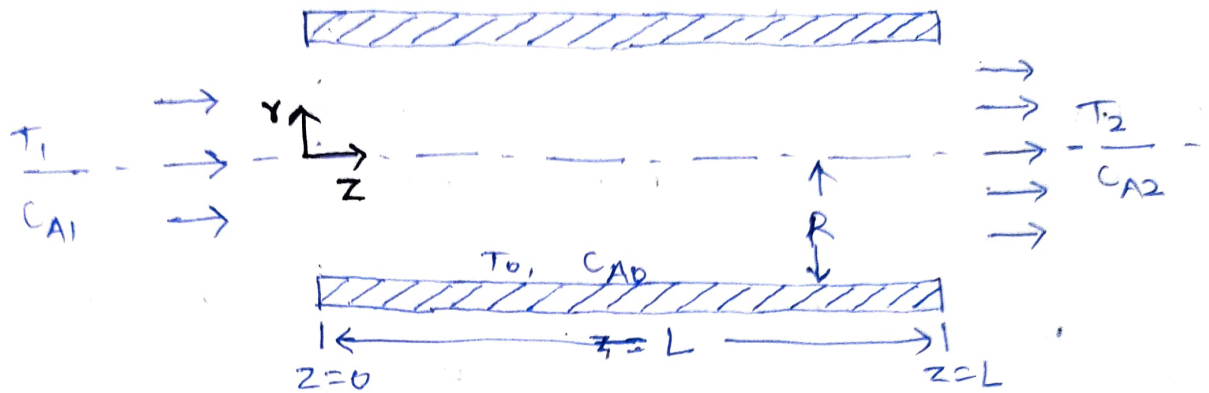
$$Sc = \frac{\text{Momentum diffusivity}}{\text{mass diffusivity}} = \frac{\nu}{D_{AB}} = \frac{\mu}{\rho D_{AB}}$$

Lewis Number

$$Le = \frac{\text{Thermal diffusivity}}{\text{mass diffusivity}} = \frac{\alpha}{D_{AB}} = \frac{k}{\rho \hat{C}_p D_{AB}}$$

correlation of Binary Transfer coefficients in one Phase at Low Mass Transfer Rates

consider Laminar/Turbulent flow of a liquid solution (A in B) through a tube.



T_1 and C_{A1} = Temperature and concentration of entering solution.

T_0 and C_{A0} = Temperature and concentration of solution at inner surface of tube.

T_2 and C_{A2} = Temperature and concentration of solution leaving the tube section.

R and L = Internal radius and length of tube

h = Heat transfer coefficient

h_m = Mass transfer coefficient

Q = Rate of Heat transfer

N_A = Rate of Mass transfer

Tube internal surface area, $A = \pi DL$ ($D = 2R$)

D_{AB} = Diffusivity of A in B

k = Thermal conductivity of tube

$T_0 > T_1$ and $C_{A0} > C_{A1}$

Assumptions:

- ① Physical properties ρ, μ, D_{AB}, k are constant.
- ② Small net mass transfer rates
- ③ No chemical reaction
- ④ No viscous dissipation heating
- ⑤ Inner surface of tube is maintained at constant temperature T_0 and constant concentration C_{A0} .

Rate of Heat addition by conduction at Tube Inner Surface equals convection through the solution:

$$Q = \int_0^L \int_0^{2\pi} \left(+k \frac{\partial T}{\partial r} \bigg|_{r=R} \right) R d\theta dz = hA(T_0 - T_1)$$

$$\Rightarrow h = \frac{1}{\pi DL(T_0 - T_1)} \int_0^L \int_0^{2\pi} \left(+k \frac{\partial T}{\partial r} \bigg|_{r=R} \right) R d\theta dz \quad (1)$$

Molar rate of addition of species A by diffusion at tube inner surface equals convection through the solution:

$$N_A = \int_0^L \int_0^{2\pi} \left(+D_{AB} \frac{\partial C_A}{\partial r} \bigg|_{r=R} \right) R d\theta dz = h_m A (C_{A0} - C_{A1})$$

$$\Rightarrow h_m = \frac{1}{\pi DL(C_{A0} - C_{A1})} \int_0^L \int_0^{2\pi} \left(+D_{AB} \frac{\partial C_A}{\partial r} \bigg|_{r=R} \right) R d\theta dz \quad (2)$$

Introducing dimensionless variables:

$$\tilde{r} = \frac{r}{D} \Rightarrow r = D\tilde{r}$$

$$\tilde{z} = \frac{z}{D} \Rightarrow z = D\tilde{z}$$

$$\tilde{T} = \frac{T - T_0}{T_1 - T_0} \Rightarrow T = (T_1 - T_0)\tilde{T} + T_0$$

$$\tilde{C}_A = \frac{\tilde{C}_A - C_{A0}}{C_{A1} - C_{A0}} \Rightarrow C_A = (C_{A1} - C_{A0}) \tilde{C}_A + C_{A0}$$

Substituting in eq. (1)

$$h = \frac{T_1 - T_0}{\pi D^2 L (T_0 - T_1)} \int_0^{L/D} \int_0^{2\pi} \left(+k \left. \frac{\partial \tilde{T}}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) \frac{D^2}{2} d\theta d\tilde{z}$$

$$\Rightarrow h = \frac{k}{2\pi L} \int_0^{L/D} \int_0^{2\pi} \left(- \left. \frac{\partial \tilde{T}}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) d\theta d\tilde{z}$$

$$\Rightarrow Nu = \frac{hD}{k} = \frac{1}{2\pi L/D} \int_0^{L/D} \int_0^{2\pi} \left(- \left. \frac{\partial \tilde{T}}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) d\theta d\tilde{z} \quad (3)$$

Non-dimensionalizing eq. (2)

$$h_m = \frac{C_{A1} - C_{A0}}{\pi D^2 L (C_{A0} - C_{A1})} \int_0^{L/D} \int_0^{2\pi} \left(+D_{AB} \left. \frac{\partial \tilde{C}_A}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) \frac{D^2}{2} d\theta d\tilde{z}$$

$$\Rightarrow h_m = \frac{D_{AB}}{2\pi L} \int_0^{L/D} \int_0^{2\pi} \left(- \left. \frac{\partial \tilde{C}_A}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) d\theta d\tilde{z}$$

$$\Rightarrow Sh = \frac{h_m D}{D_{AB}} = \frac{1}{2\pi L/D} \int_0^{L/D} \int_0^{2\pi} \left(- \left. \frac{\partial \tilde{C}_A}{\partial \tilde{r}} \right|_{\tilde{r}=\frac{1}{2}} \right) d\theta d\tilde{z} \quad (4)$$

Solution of Non-dimensional form of Equation of continuity, ~~motion~~ Equation of Motion and Equation of Energy gives:

$$\tilde{T} = f_1(\tilde{r}, \theta, \tilde{z}, Re, Pr) \quad \& \quad \tilde{C}_A = f_1(\tilde{r}, \theta, \tilde{z}, Re, Sc)$$

Substituting in eq. (3) and (4) gives

$$Nu = f_2(Re, Pr, \frac{L}{D}) \quad \& \quad Sh = f_2(Re, Sc, \frac{L}{D})$$

Stanton number for Heat transfer

$$St = \frac{Nu}{Re \cdot Pr}$$

$$Re = \frac{DU}{\nu}$$

$$Pr = \frac{\nu}{\alpha}$$

Stanton number for mass transfer

$$St_m = \frac{Sh}{Re \cdot Sc}$$

$$Sc = \frac{\nu}{D_{AB}}$$

Reynolds Analogy:

$$\frac{f}{2} = St = St_m$$

CHILTON, COLBURN ANALOGY

This is a modification of Reynolds Analogy

correction factor $Pr^{2/3}$ is added to Reynolds analogy for heat transfer

$$\frac{\bar{C}_F}{2} = St \cdot Pr^{2/3} = J_H$$

$$0.6 < Pr < 60$$

$$\frac{\bar{C}_F}{2} = St_m \cdot Sc^{2/3} = J_m$$

$$0.6 < Sc < 300$$

correction factor $Sc^{2/3}$ is added to Reynolds analogy for mass transfer.

$$\Rightarrow \frac{\bar{C}_F}{2} = St \cdot Pr^{2/3} = St_m \cdot Sc^{2/3}$$

$$\Rightarrow \frac{\bar{C}_F}{2} = J_H = J_m$$

Summary of Analogies

① Reynolds

$$St = \frac{C_f}{2} = \frac{f}{2}$$

$$St_m = \frac{C_f}{2} = \frac{f}{2}$$

($C_f = f = \text{friction factor}$)

② Chilton-Colburn

$$St = \frac{f}{2} Pr^{-2/3}$$

$$St_m = \frac{f}{2} Sc^{-2/3}$$

③ Prandtl

$$St = \frac{f/2}{1 + 5\sqrt{\frac{f}{2}}(Pr-1)} = \frac{f/2}{1 + 5(Pr-1)\sqrt{\frac{f}{2}}}$$

$$St_m = \frac{f/2}{1 + 5(Sc-1)\sqrt{\frac{f}{2}}}$$

④ Martinelli

$$St = \frac{f/2}{1 + 5 \left[Pr - 1 + \ln\left(\frac{1 + 5Pr}{6}\right) \right] \sqrt{\frac{f}{2}}}$$

$$St_m = \frac{f/2}{1 + 5 \left[Sc - 1 + \ln\left(\frac{1 + 5Sc}{6}\right) \right] \sqrt{\frac{f}{2}}}$$

Q. A spherical water droplet 0.05 cm in diameter is falling at velocity of 215 cm/s through dry still air at 1 atm pressure. Estimate instantaneous rate of evaporation from the drop if drop surface is at 21°C and air at 60°C. Vapor pressure of water at 21°C is 0.0247 atm. Assume pseudo steady state condition and $k_{am} = 1.35 \times 10^{-3} \text{ mol s}^{-1} \text{ cm}^{-2}$.

Soln: Given,

Diameter of droplet, $D = 0.05 \text{ cm}$

velocity, $U = 215 \text{ cm/s}$

Total pressure, $P = 1 \text{ atm}$

Temperature of drop surface, $T_0 = 21^\circ\text{C}$

Temperature of bulk air, $T_\infty = 60^\circ\text{C}$

Vapor pressure of water at 21°C = 0.0247 atm

$k_{am} = 1.35 \times 10^{-3} \text{ mol s}^{-1} \text{ cm}^{-2}$

Let. $A \rightarrow \text{water}$

$B \rightarrow \text{air}$

mol fraction of A at AB interface

$$x_{A0} = \frac{P_A}{P} = \frac{0.0247}{1} = 0.0247$$

mol fraction of A in bulk air, $x_{A\infty} = 0$ (dry air)

At droplet surface (AB interface):

$$N_{A0} - x_{A0}(N_{A0} + N_{B0}) = -c D_{AB} \left. \frac{dx_A}{dr} \right|_{r=R} = k_{xm}(x_{A0} - x_{A\infty})$$

where, k_{xm} = Mean value of mass transfer coefficient based on mole fraction

N_i = Molar flux

As B is non-diffusing, $N_{B0} = 0$.

molar rate $W_i = A N_i$

where, $A = \pi D^2$ (surface area)

$$(1 - x_{A0}) W_{A0} = k_{xm} A (x_{A0} - x_{A\infty})$$

$$\Rightarrow W_{A0} = k_{xm} \pi D^2 \left(\frac{x_{A0} - x_{A\infty}}{1 - x_{A0}} \right)$$

$$= k_{xm} \pi D^2 \left(\frac{x_{A0}}{1 - x_{A0}} \right)$$

$$= 1.35 \times 10^{-3} \times 0.05^2 \pi \left(\frac{0.0247}{1 - 0.0247} \right)$$

$$= 2.7 \times 10^{-7} \text{ gmol/s}$$